BABU BANARASI DAS UNIVERSITY

LUCKNOW

Session : 2026-2027

SCHOOL OF COMPUTER APPLICATIONS

PRACTICAL FILE

(BCADSN25301)

Submitted By:-

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Section – BCADS31
Semester – 5th
Program – BCA (DS & AI)
Course Name – Predictive Analytics
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Trainer:-

Mr. Karan Raj
(IBM MSME)

Project - 2

Definition: - This practical demonstrates how to **prepare customer data, build a CHAID model, predict response, and select customers likely to buy the product**.

Outcomes/Learning: - Learners will understand how to **prepare data for modeling, build a CHAID model, interpret predictions, and identify potential customers for a campaign**.

Purpose: - To build a model that predicts which customers are likely to **respond positively to an ACME campaign**.

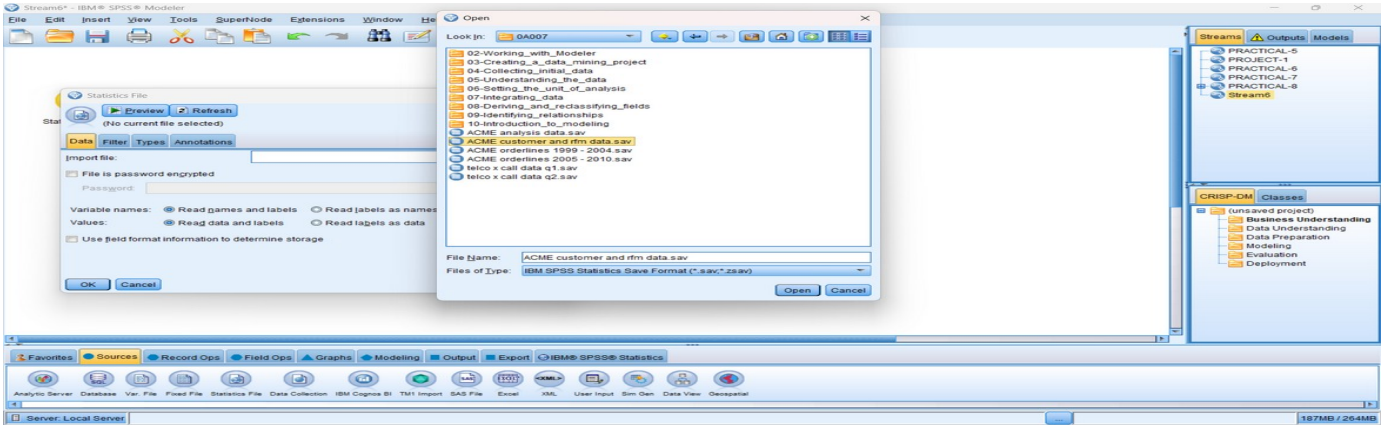
TASK – 1

❖ Start IBM SPSS Modeler and set the working folder and Import and Examine the Data.

Definition: - This task explains how to **start IBM SPSS Modeler, set the working folder,** and how to **import the telecommunications modeling data and examine its fields before building a model.**

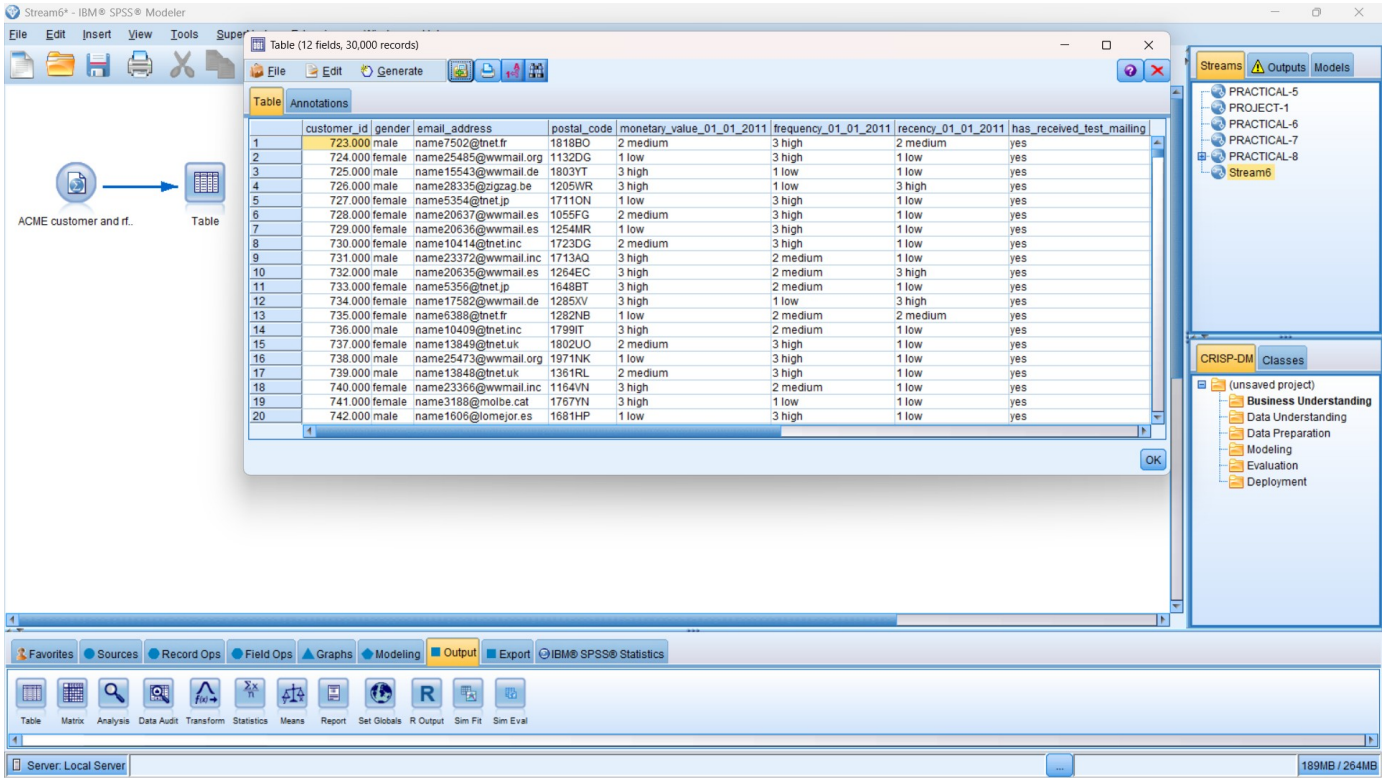
Step 1: First of all, from the File menu, click Set Directory and choose your path.

- Beside Look in, navigate to **C:\Users\Mr. Ajay\OneDrive\Desktop\SPSS_DEMO\Data Set PA\0A007**, and then click **Set**.
- From the **Sources** palette, place a **Statistics File** node on the stream canvas.
- Double-click on the **Excel** node and in this **Excel** node.
- Like: **C:\Users\Mr. Ajay\OneDrive\Desktop\SPSS_DEMO\Data Set PA\0A007\ACME customer and rfm data.sav**



- Click **"Open"**, and then click **"OK"** to close the **Statistics** dialog box.

Step 2: From the **Output** palette, add a **Table** node downstream from the **Statistics File** node.



- Click **"OK"** to close the Table output window

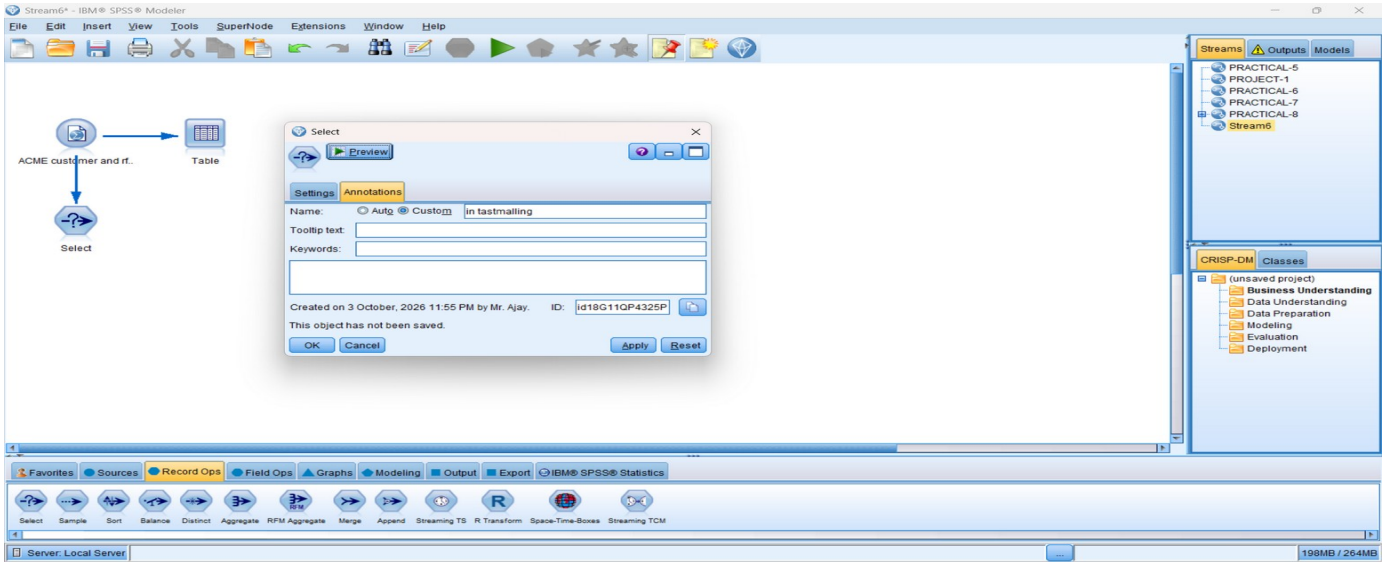
TASK – 2

❖ Select the Data to Build the Model

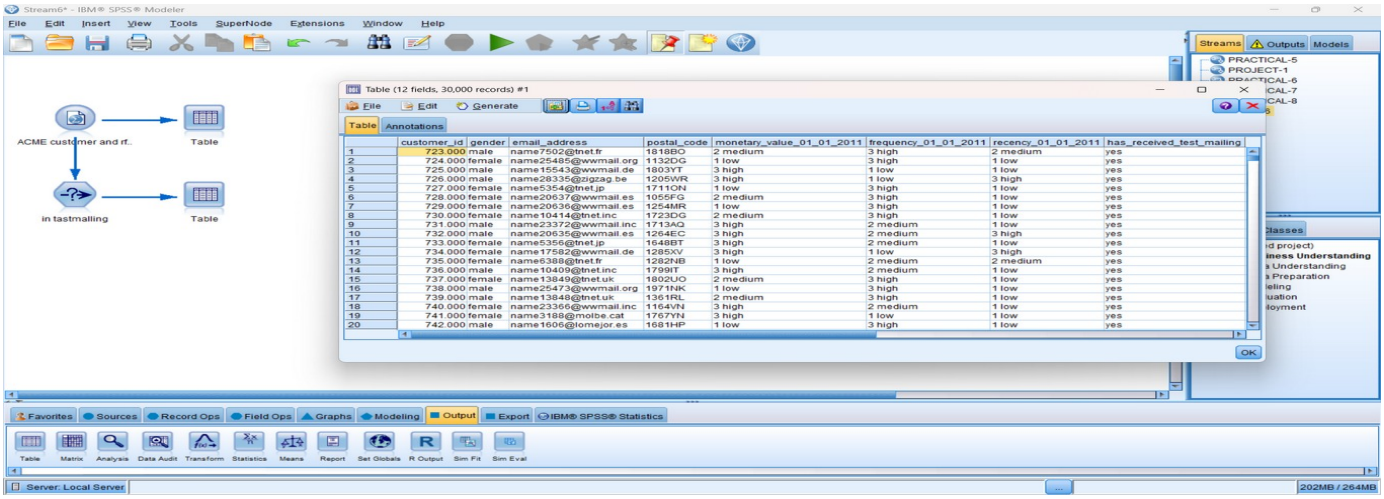
Definition: - This task demonstrates how to **select only the customers who were included in the mailing test**

Step 1: For this:

- From the **Record Ops** palette, add a **Select** node downstream from the **Statistics File** node.
- Double-click on the **Select** node, in the condition box write **has_received_test_mailing = "yes"**; then click Annotations. In the Name field, click Custom, rename it to tastmalling, then click **"OK".**



Step 2: From the **Output** palette, add a **Table** node downstream from the **Select** node.



- Click **"OK"** to close the Table output window.

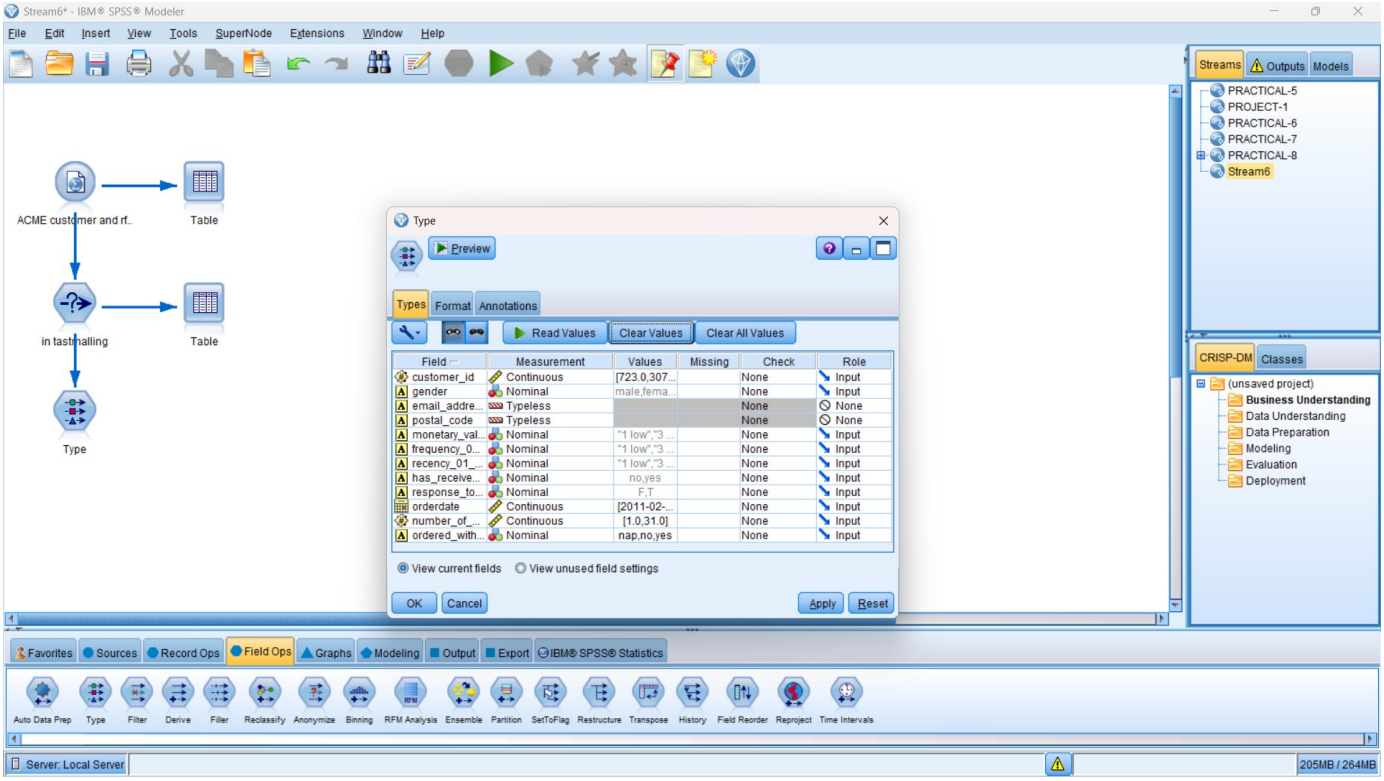
TASK – 3

❖ Build a CHAID Model

Definition: - This task demonstrates how to **build a CHAID model to predict customers response using RFM and gender fields**.

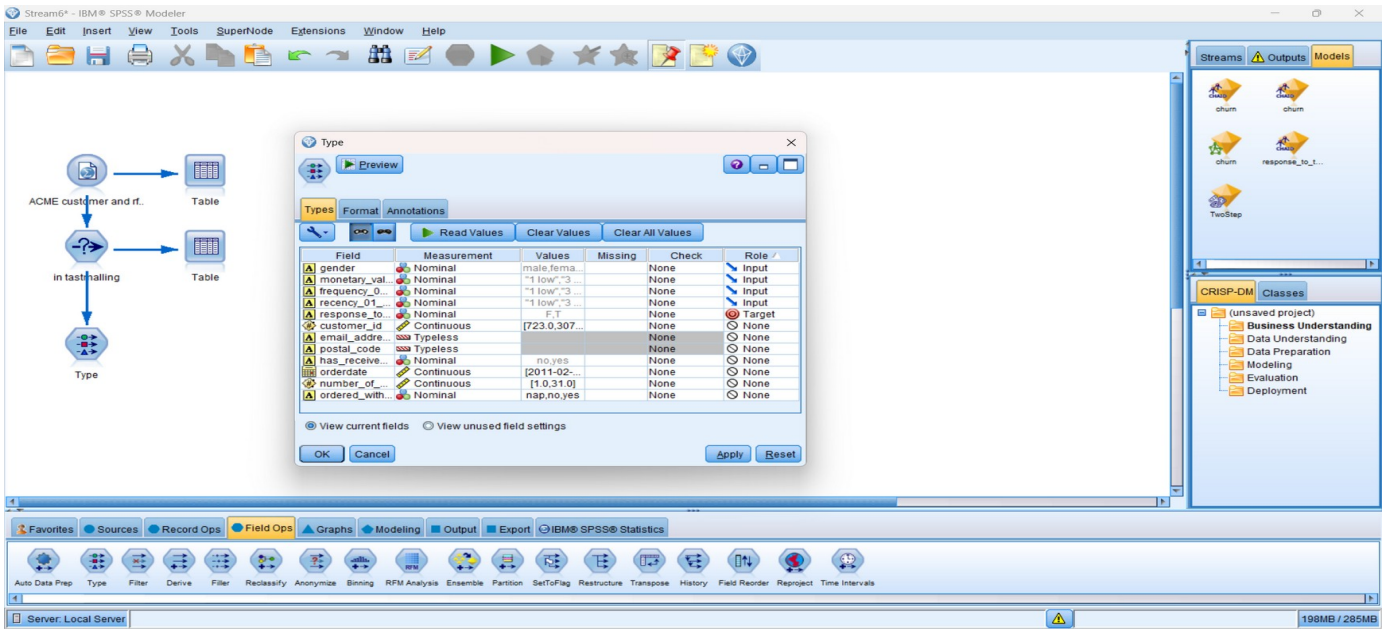
Step 1: For this:

- From the **Field Ops** palette, add a **Type** node downstream from the **Select** node.
- Double-click on the **Type** node, then click **Read Values**.



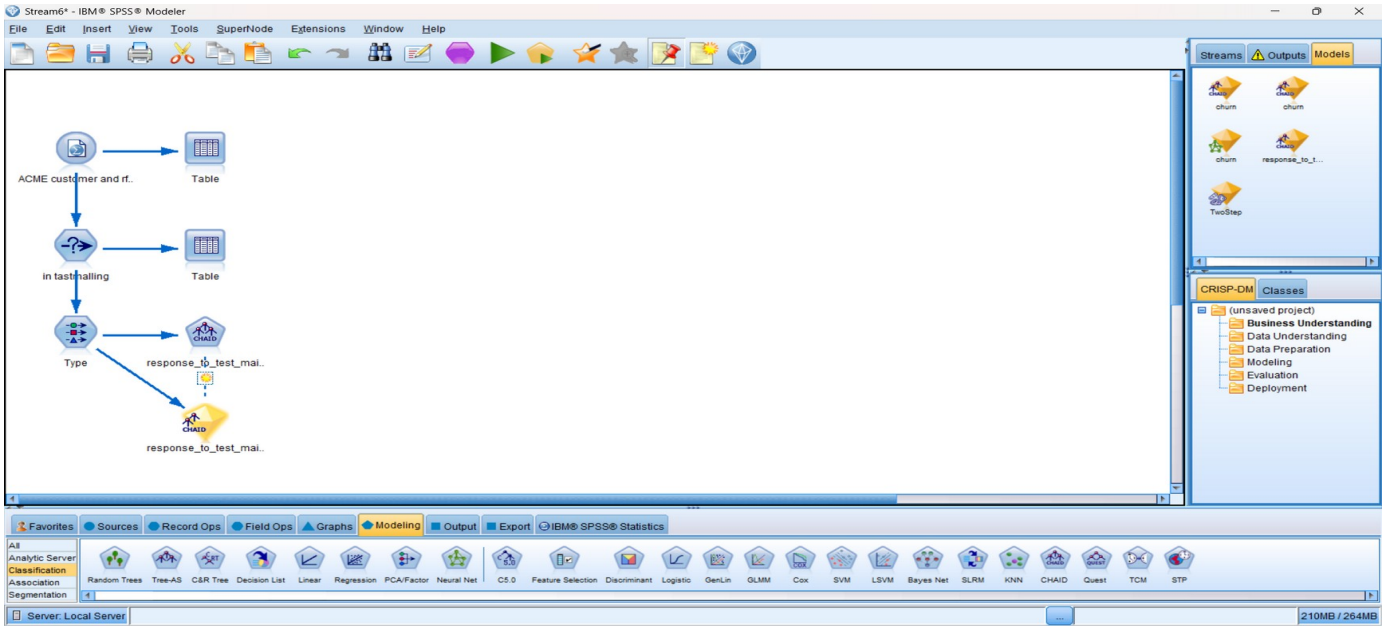
Step 2: Set these fields as **Input**:

- **Gender**
- **monetary_value_01_01_2011**
- **frequency_01_01_2011**
- **recency_01_01_2011**
- Set **response** as the **Target**. And set other fields to **None**. After that, sort the **Role.**



- Click **"OK"** to close the **Type** dialogue box.

Step 3: From the **Modeling** palette, on the left side, click **Classification** and add a **CHAID** node downstream from the **Type** node. Then run the **CHAID** node.

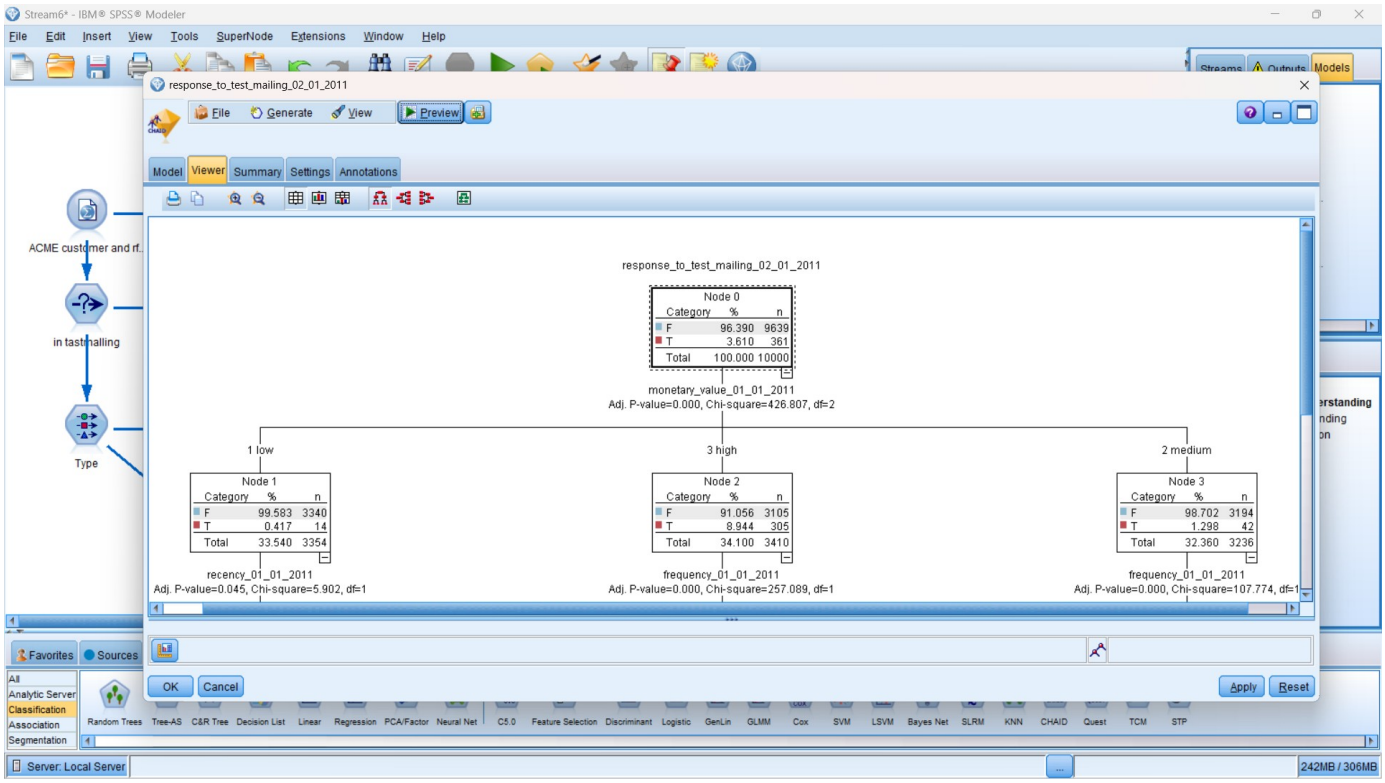


TASK – 4

❖ Build a CHAID Model

Definition: - This task demonstrates how to **examine the CHAID model and identify the customer group with the highest response rate**.

Step 1: Double-click the **CHAID model nugget.** Then click the **Viewer** tab.



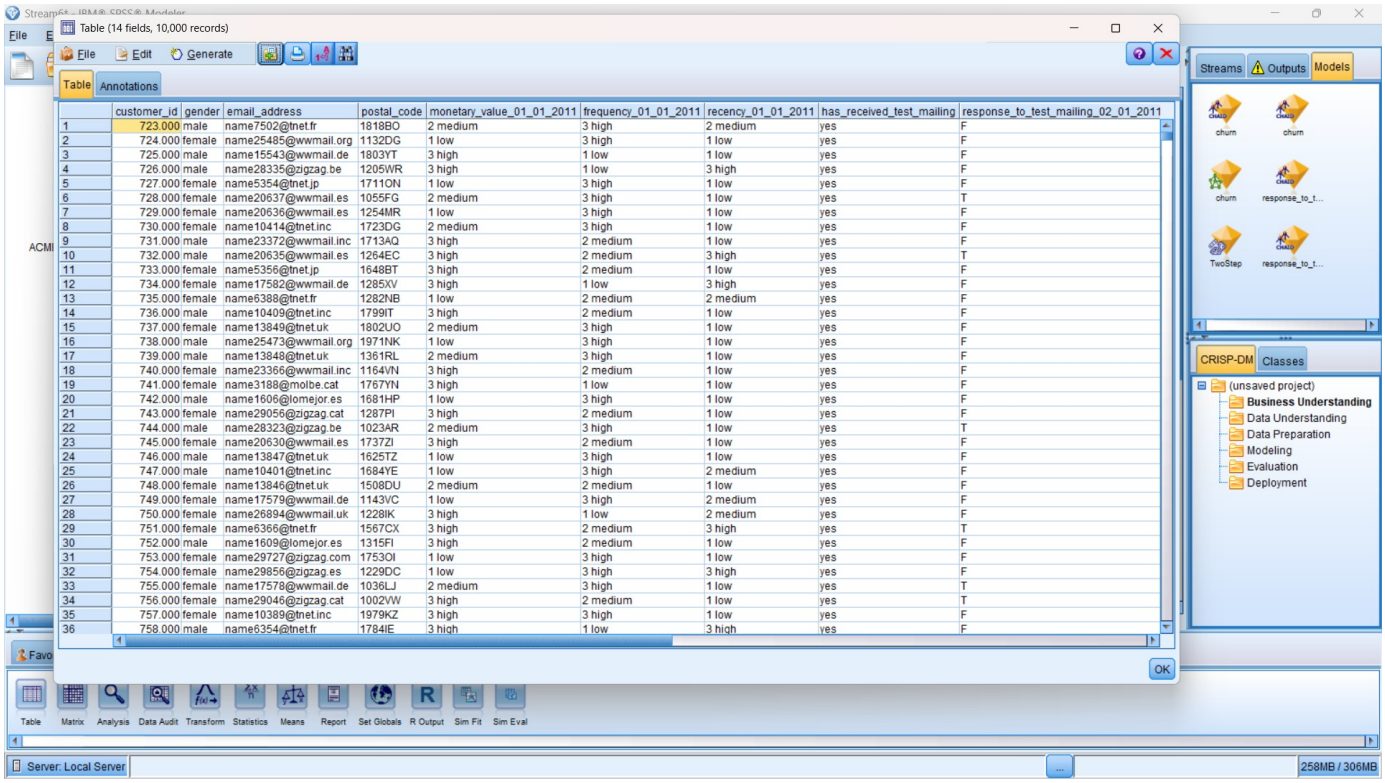
- Locate the group with the highest response rate.
- The highest-response group has **high monetary value, medium or high frequency, and high recency**, with a response probability of **0.625**.
- Click **"OK"** to close the dialogue box.

TASK – 5

❖ Interpret the Fields Added by the Model Nugget

Definition: - This task demonstrates how to **interpret the predicted response and confidence fields generated by the model**.

Step 1: From the **Output** palette, add a **Type** node downstream from the model nugget. Then run the table node.



- Click **"OK"** to close the **Table** dialogue box.

TASK – 6

❖ Use the Model to Score the Rest of the Customers

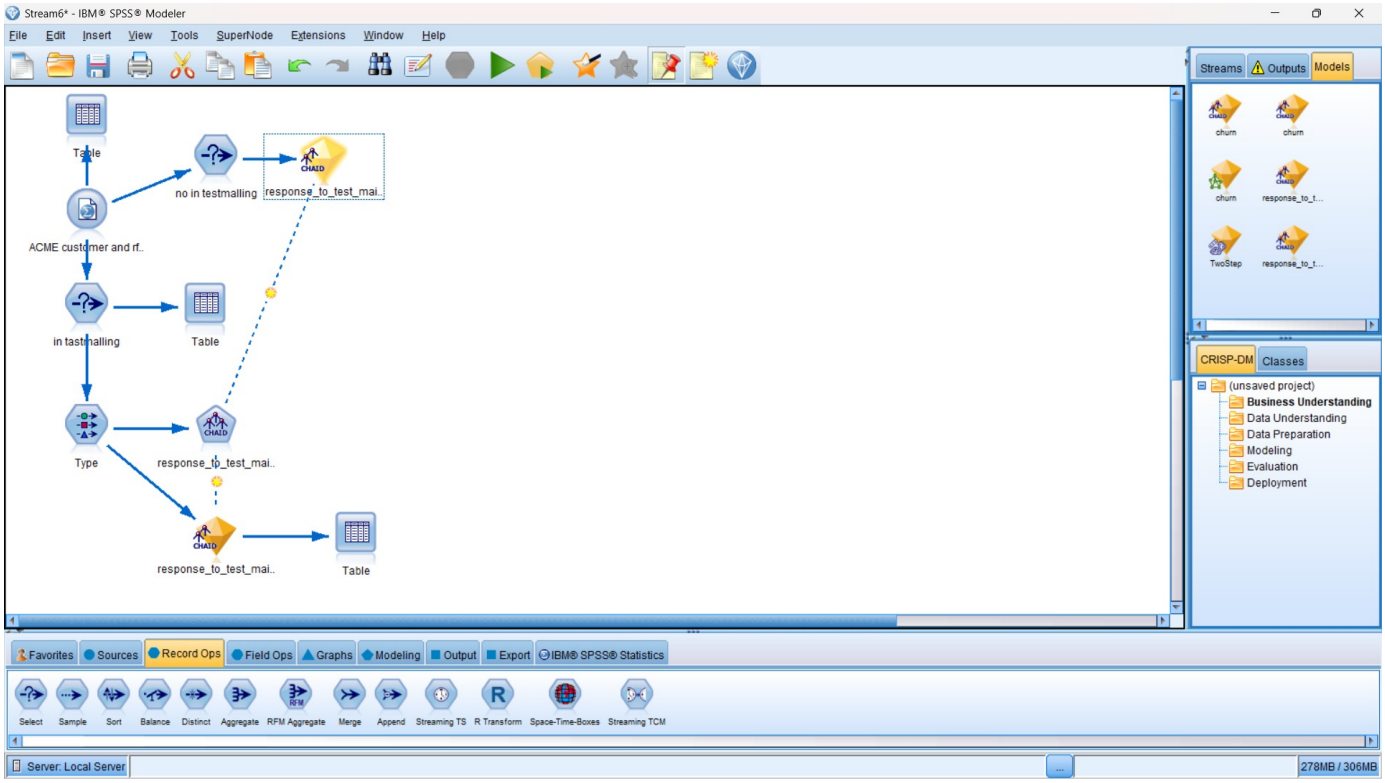
Definition: - This task demonstrates how to **apply the trained model to customers who we're not included in the test mailing**.

Step 1: For this:

- From the **Record Ops** palette, add a **Select** node downstream from the **Statistics File** node.
- Double-click on the **Select** node, in the condition box write

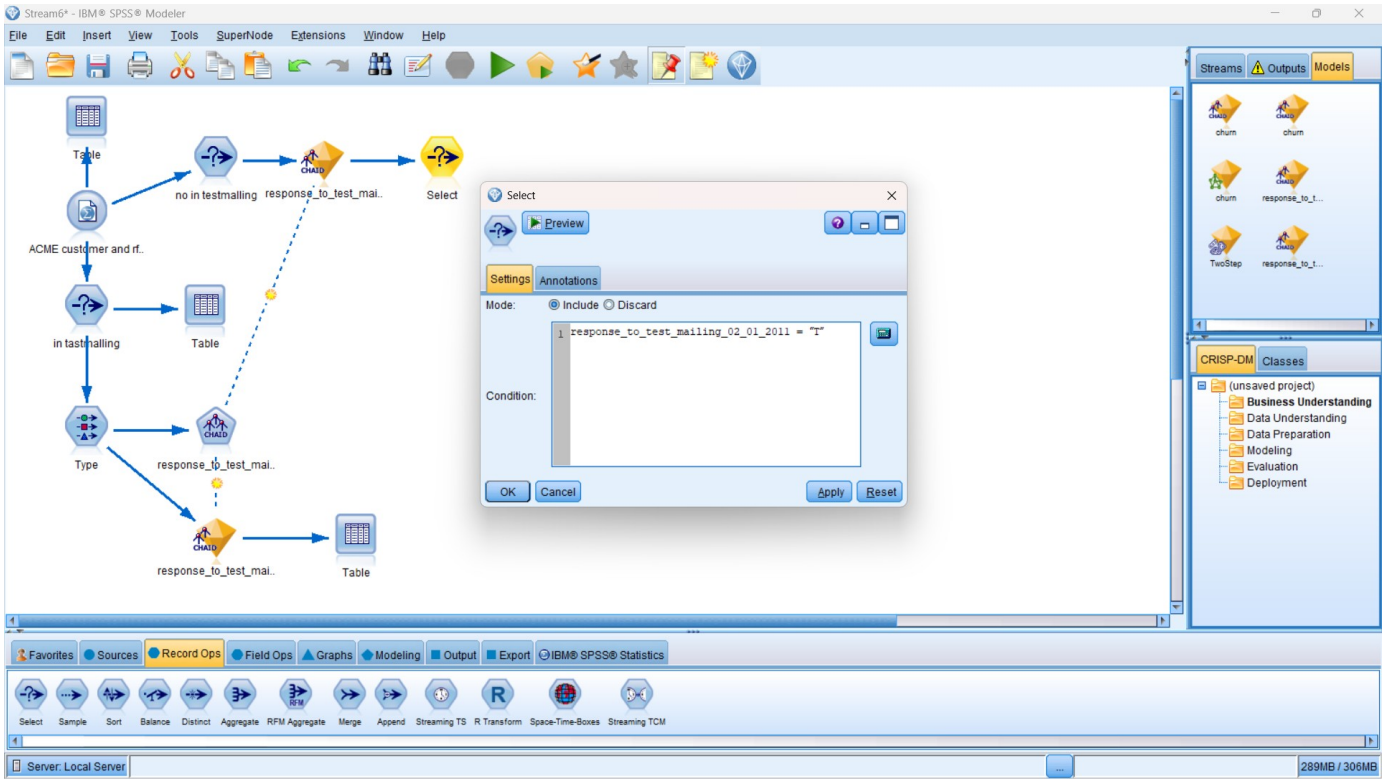
has_received_test_mailing = "no" then click Annotations.

- In the Name field, click Custom, rename it to no in tastmalling, then click **"OK".**
- After that, copy and paste the **CHAID model nugget** downstream from the Select node.



<u>Step 2:</u> For this:

- From the **Record Ops** palette, add a **Select** node downstream from the **CHAID model nugget** node.
- Double-click on the **Select** node; in the condition box, write **response_to_test_mailing_02_01_2011 = "T"** then click Annotations. In the Name field, click Custom, rename it to **predicted to buy**, then click **"OK".**



• Run a **Table** node to get the record count; 254 customers are predicted to buy the product.

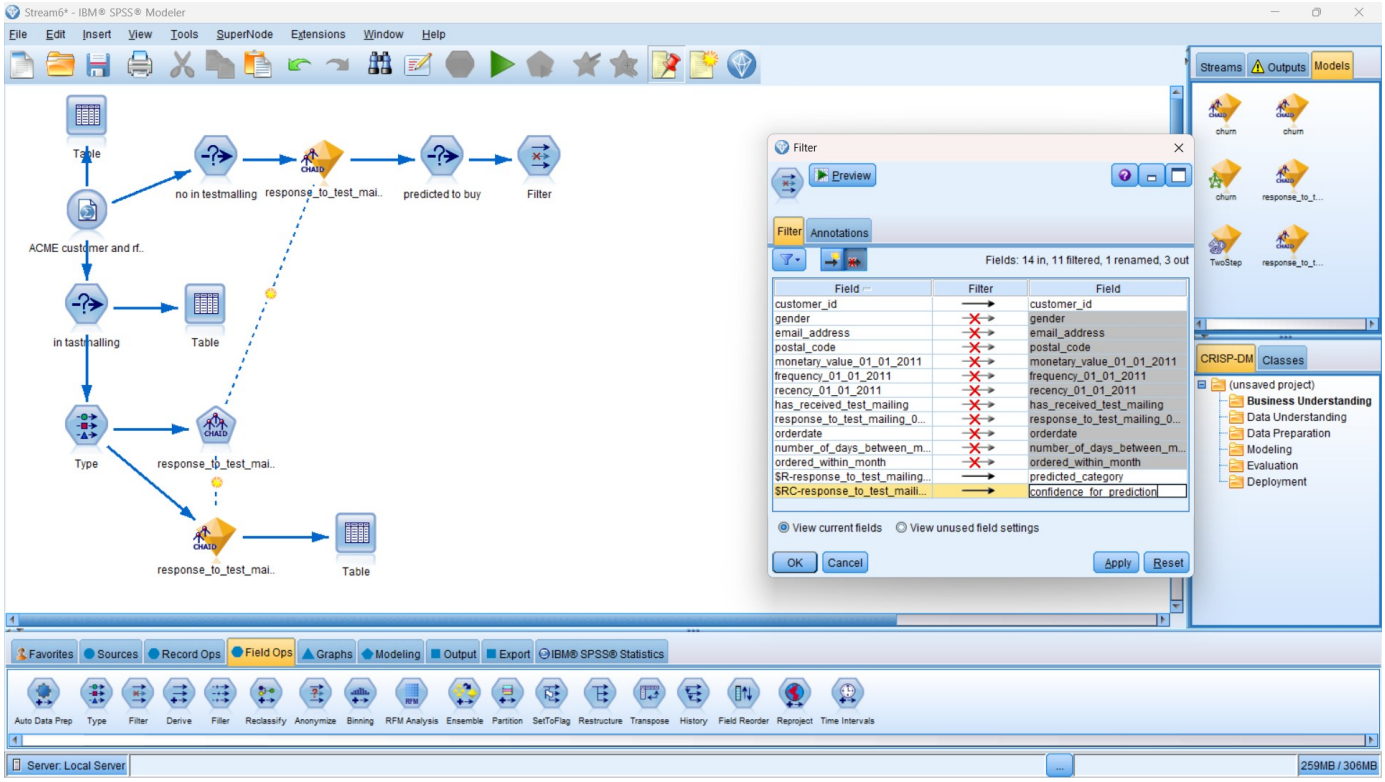
TASK – 7

❖ Export the Results

Definition: - This task demonstrates how to **filter the selected customers and export their details to a text file**.

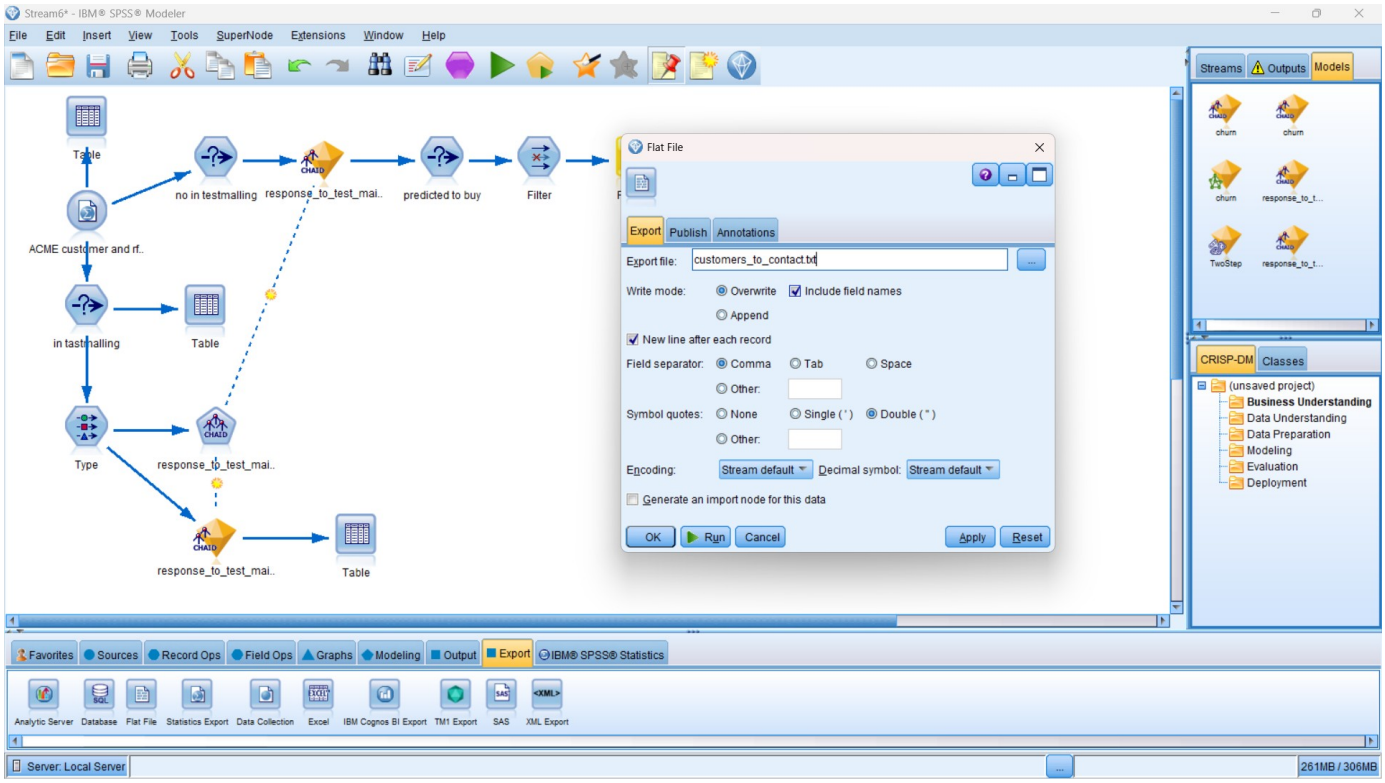
Step 1: For this:

- From the **Filter Ops** palette, add a **Select** node downstream from the **predicted to buy** node.
- Click on the icon to disable all the.
- Keep only:
    - **customer_id**
    - **$R-response**
    - **$RC-response**
- Rename the fields:
    - **$R-response_to_text… → predicted_category**
    - **$RC-response_to_text… → confidence_for_prediction**



- Click **"OK",** to close the filter box.

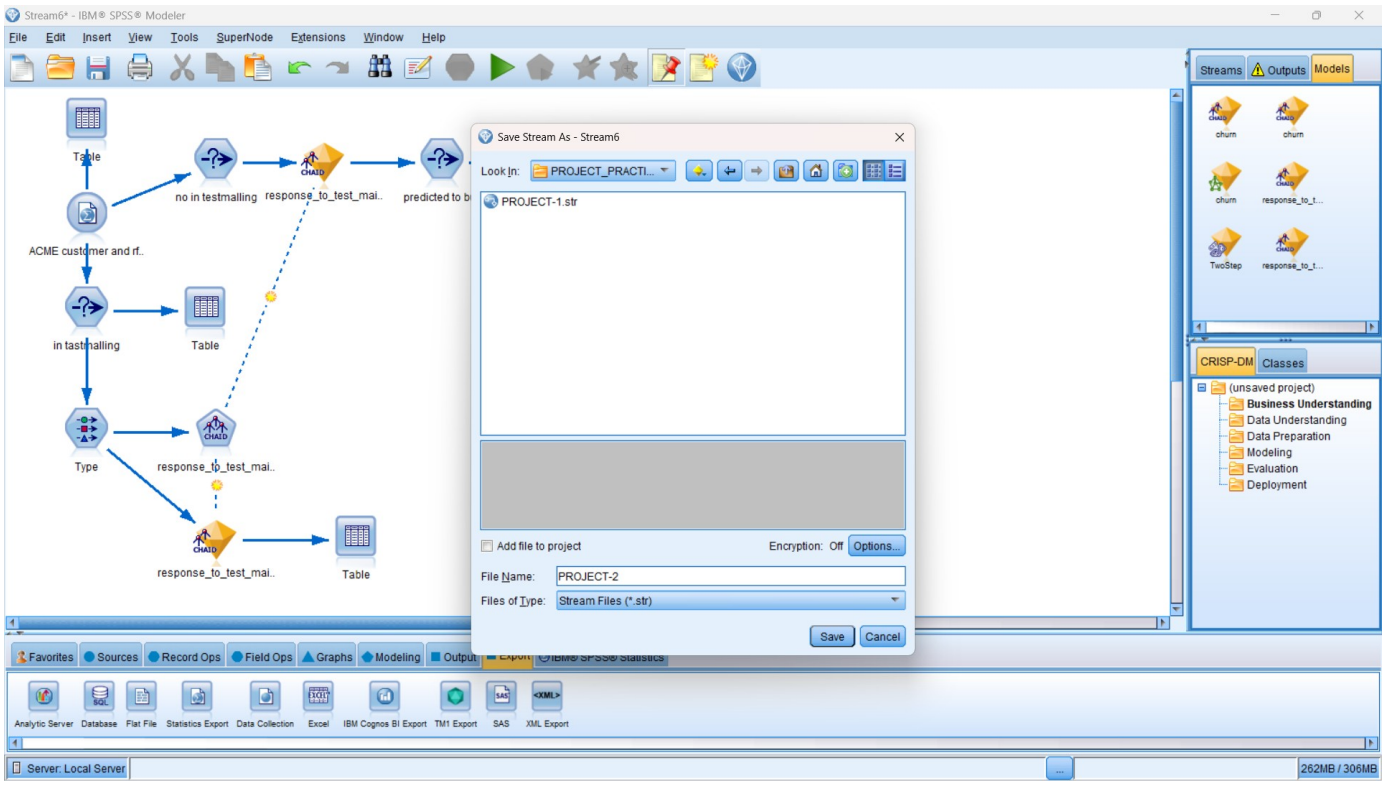
<u>Step 2:</u> From the **Export** palette, add a **Flate File** node downstream from the **Filter** node.
Double-click on the **Flate File** node; in the Export file box, write
customers_to_contact.txt.
And run the **Flat File** node to export the data.



Step 3: To save the stream, go to the File menu and click **Save** or **press Ctrl+S**. Choose the desired location/path.

Like: - **C:\Users\Mr. Ajay\OneDrive\Desktop\SPSS_DEMOC: \Users\Mr. Ajay\OneDrive\Desktop\SPSS_DEMO\PROJECT_PRACTICAL**

Enter the file name you want, and then click **Save**.



Result: The **CHAID model** was successfully built to predict customer response, applied to the remaining customers, and the customers predicted to buy were identified and exported for further use.